

The impact of regulation on market risk[☆]

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Abstract

While it is crucial to understand the impact of regulatory changes on market risk, the literature does not show how risk responds to expected regulatory changes that are specifically designed to change risk. Our paper fills this gap by providing a detailed study of one such case. Using both a sample of privatized U.K. companies, and U.K. and U.S. control portfolios, between 1993 and 2000, we show (both for the single-factor market model and the three-factor Fama-French model) that the observed changes in market risk are significant and consistent with theory.

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1. Introduction

The direct and indirect regulation of quoted companies is a common and growing feature of “stock market” economies. These regulations are usually intended to impact on the returns and/or risk of those holding claims on the underlying assets. For example, competition law, price controls, investor protection, market entry restrictions, etc., are all

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intended to facilitate the transfer of wealth between parties either as an end in its own right or as a means to achieve specific objectives. A significant literature documents the effect of regulation on valuation.¹ In contrast, no equivalent literature shows how risk responds to expected regulatory changes that are specifically designed to impact on risk. The scarcity of such evidence is not surprising, however, because while regulatory changes typically impact on risk, there is a lack of relatively “clean” cases that isolate the risk effect. That is, very few regulatory changes that apply to quoted nonfinancial institutions are designed primarily to transfer risk but not wealth between parties, and hence few regulatory changes enable the risk effect to be isolated for investigation.

This paper addresses the above gap in the literature. We provide a detailed study of a case in which regulation is introduced specifically to transfer risk between parties and not to create a wealth transfer. We inquire whether the market risk response is significant and, if so, whether it is consistent with theory. The paper analyzes the time-varying nature of risk and shows (using a single-factor and three-factor Fama-French model) that the impact is both significant and consistent with the theoretical predictions.

The question of how market risk responds to regulatory changes designed to change risk is now more crucial in international markets than ever before due to the scale of the activities involved. For instance, the proportion of internationally diversified portfolios that is affected by changes in regulation is huge and growing. Moreover, a wave of privatization has taken place around the world over the last twenty years. May 2003 estimates put the total value of assets transferred from the public to the private sector at \$3.24 trillion, which is 18.2% of the global stock market value and 38.6% of the non-U.S. total value (Megginson, 2005, Chapter 7). Many of these companies (e.g., telecommunications, electricity, gas markets) have been subject to some form of specific regulatory control (see, for example, Dewenter and Malatesta, 1997; Grout et al., 2004; Jones et al., 1999). Even within well-developed economies, regulated privatized companies now account for a significant fraction of the stock market (e.g., at least 13.1% in Germany, 11.7% in Australia, and 7.7% in France (see Megginson and Netter, 2001) with many privatizations still to come (see also Boutchkova and Megginson, 2000, for a study on the impact of share issue privatizations on the growth of stock markets). In addition, where there has been little utility privatization (e.g., the U.S.), existing regulation of utilities is unlikely to decline dramatically, with the exception of telecommunications. Furthermore, the market provision of more traditional public services is growing in the U.S. and elsewhere. For example, Lopez-de-Silanes et al. (1997) show that in the U.S. newly provided services of this type are 50% more likely to be provided by private contractors than by county governments. Indeed, in some countries, the extreme example being New Zealand, the regulation of privatized companies is increasing, rather than declining, as companies move further from privatization. Taken together, the above results suggest that globally, the direct economic regulation of privately owned companies appears set to continue on an upward trend.

¹See, e.g., Bittlingmayer and Hazlett (2000), Eckel et al. (1997), Fields et al. (1990), and Prager (1989). Kole and Lehn (1999) discuss the relationship between deregulation, market performance and corporate structure of the U.S. airline industry. Binder (1985) discusses the problem of undertaking event studies of regulation changes (see Section 3 below). Using an international panel from the 1990s La Porta et al. (2002) show that valuation is higher where minority shareholders have stronger protection.

The results of this paper are also important because financial markets, not just stock markets, are the primary source of finance for quoted companies. For example, Wurgler (2000), and Beck et al. (2000) show that financial markets, rather than state ownership, improve the allocation of capital. Companies that face significant direct and indirect regulation fall between the state ownership and fully private models. How regulation affects the risk of these companies is therefore important both for portfolio choice and for policy design. Of course, the ability of regulators to intervene at any time, i.e., the incomplete nature of the contract, may affect investment in important ways other than changing the cost of capital (see, for example, Grout, 1984; Hart, 1995). This is not the concern of the paper here, however. The above discussion emphasizes the positive aspect of regulation, i.e., the role of regulation in improving the efficient operation of markets. There is no guarantee, however, that most regulation exists for this purpose. The primary reason for a given regulation's existence could be the benefit it bestows on politicians and bureaucrats (see, for example, the “grabbing hand” analogies in Djankov et al., 2002). Even in such a case, however, how the markets react is still just as valid a design requirement, albeit without an efficiency justification.

The particular case study of this paper concerns the regulation of telecommunications, water, electricity, and airports in the U.K. in the 1990s. In the late 1990s the U.K. government proposed a change in the pricing system of all regulated U.K. utilities, from price-cap to explicit profit sharing between companies and customers. The shift to profit sharing (whereby companies share profit increases and decreases with customers) is an excellent example of a “clean” case that isolates the risk effect. As far as we are aware this is the only example of a universal (i.e., covering all regulated sectors) proposed change to profit sharing from a different regulatory regime. After a period of 25 months (here in referred to as the *profit sharing period*) the government abandoned the plan. In the paper we analyze the time-varying pattern of risk for each of the companies affected from 1993 to 2000 using both a single-factor model and the Fama-French three-factor (FF3F) model. The theoretical prediction for the impact on these companies is that the risk of the regulated companies should be lower in this 25-month period relative to a control portfolio. Applying Kalman filter techniques, OLS with White heteroskedasticity-consistent standard errors and covariance corrections (OLS WHC), and ML with GARCH(1,1) effects to a single-factor market model and the FF3F model, we show that this effect is strongly confirmed in the data.

Using daily data with a market model, the effect in the profit sharing period is very clear on time-varying betas for a portfolio of regulated companies. This is true whether we apply the Kalman filter technique to the full sample period or to three subsamples (before, during, and after the treatment). The effect is also clear when we apply the Kalman filter to individual companies. Applying OLS WHC and ML with a GARCH(1,1) specification, we find that the effect on risk (as measured by beta) is significant at the 1% level on a portfolio of regulated companies and on almost every company at the individual company level.

To show that the regulatory-period changes in risk are U.K. specific, we compare U.S. and U.K. fluctuations in market risk for corresponding groups of companies. Choosing the U.S. market as a comparator helps us control for changes in demand and supply conditions (worldwide) that are unrelated to the regulation of the ownership structure inside the U.K. We show that there is no change in market risk for the U.S. portfolios during the period of the regulatory debate in the U.K. We also argue that changes in leverage cannot explain the observed decline in market risk. However, we find a significant

decline in risk in both U.S. portfolios in the mid-1999 to end-2000 period. This decline is consistent with the pattern observed for the U.K. market for all portfolios studied. Although not central for the core result in the paper (since the test depends on the differences between portfolios), we conjecture on the cause of this decline (the e-commerce “bubble”) and provide evidence that explains why we observe a significant effect.

We also investigate the effect of the regulatory change on risk using the FF3F model on monthly data. While the effect is confirmed, the impact of the two additional factors is less clear than that of the market factor. Indeed, the Kalman filter regressions confirm that the effect is significant through the market factor. When we apply OLS WHC to the portfolios, the profit sharing dummies are significant for all three factors. This is true, although far weaker, when we address data at the individual company level.

Here, the study of the regulatory change’s impact on systematic risk is the primary question under investigation. However, the case study, also provides a tangential but intriguing contribution to the debate on the role of the FF3F. Typically studies that assess whether additional, i.e., non-CAPM, factors are significant address the extent to which they can explain returns over time across many companies. In contrast, in this paper we are concerned with picking up a predicted effect in an experiment. We are able to identify, from government publications and actions, a specific proposed policy change to a specific set of companies for a specific period of time. We are able to predict from theory what effect this should have on systematic risk. Taking the investigation as a whole there is very convincing evidence that the effect is strong and in the direction predicted. This, then, can be used as an informative experiment to throw light on the relevance of various factors. We find that all three FF factors are significant in picking up the predicted effect, although the HML factor is only significant at the 10% level.

The organization of the paper is as follows. Section 2 provides a theoretical presentation of the predicted risk-regulation relation for different regulatory regimes discussed in our study. Section 3 describes and discusses the data. We construct a sample of 15 privatized regulated companies and a control sample of 18 unregulated companies that will form the core of our analysis. Section 4 tests our theoretical predictions using a single-factor market model and gives the associated estimation results using the Kalman filter approach with a GARCH(1,1) specification, OLS WHC, and ML with a GARCH(1,1) specification for the portfolios constructed in Section 3. Section 5 gives the comparison of the Kalman filter estimates obtained for the U.K. and U.S. markets based on portfolios created from DataStream sector indexes that contain companies selected in Section 3. This section also discusses leverage effects. Section 6 applies the Kalman filter and OLS WHC estimation techniques to a FF3F model. Section 7 closes with conclusions.

2. Regulation and risk

2.1. Risk and regulation

Historically, rate-of-return regulation has been the most common form of direct price control regulation for companies. The central feature of a rate-of-return regime is that prices are adjusted to ensure that the rate of return on the company’s assets does not exceed some specified level. As a result the company does not benefit from achieving lower costs and the consequences of inefficiency are more easily passed on to consumers. These incentives are in stark contrast to those that arise in a competitive market, where gains that

a company is able to make, e.g., from implementing a new, cheaper technology, will lead to higher profit for a period of time. Of course, the innovator's benefit will be eroded over time as competitors catch up. The competitive process will push prices down, passing the benefits on to the market, and will culminate in a position in which the innovator again earns no greater reward than others in the industry. The failure of rate-of-return regulation to replicate the risk and rewards inherent in competition have led to the well-documented result that it provides poor incentives for efficiency improvements, leading to higher costs and overinvestment (see [Armstrong et al., 1994](#), for a thorough discussion of models of regulation).

In contrast, price-cap regulation sets a pre-determined ceiling on prices for a number of years. During this time the ceiling is only allowed to increase by the rate of inflation less some pre-determined adjustment factor. For example, in U.K. telecommunications, the first U.K. utility sector to be privatized, the adjustment factor is called X and for this reason price-cap regulation is frequently referred to as RPI-X regulation. The adjustment factor is set at the start of the five-year period at a level that will enable the regulated entity to earn an expected fair rate of return on its assets, provided it is run efficiently. If, through greater effort, the regulated company is able to reduce costs at a greater rate than expected, or if business conditions are better than expected, prices still follow the pre-specified rule and the company will earn an even higher return for the life of the price-cap. Similarly, if the company is less efficient or faces unexpected adverse business conditions, the company earns a lower return than expected. The effect is that the regulated entity is able to retain the rewards of greater effort for a few years but these are passed on to the consumer at the next price review. Overall, price-cap regulation aims to mimic the benefits of the competitive process, bringing to the regulated firm the risks and the rewards that would arise in a competitive industry.²

Profit sharing is a compromise form of regulation between rate-of-return and price-cap regulation. Under profit sharing the regulated firm's profits and losses are explicitly shared with consumers according to some pre-specified rule. Each of these models of regulation implies different risks for a regulated company. This is considered in the next subsection.

2.2. Risk and profit sharing

This subsection provides a simple theoretical illustration of the impact that the regulatory change should have on a company's risk. If the firm faces profit sharing regulation, then its profits, $\hat{\pi}$, can be represented as a restricted deviation around a target profit level, π^* . More formally,

$$\hat{\pi} = \pi^* + \gamma(\tilde{\pi} - \pi^*),$$

where $\gamma < 1$ is the proportion the firm is allowed to keep if its profit exceeds the target profit π^* , and $\tilde{\pi}$ is the profit the company would earn under price-cap regulation.

The above formula can be rewritten as a weighted average of the outcome under rate-of-return regulation, π^* , and the outcome under price-cap regulation, $\tilde{\pi}$:

$$\hat{\pi} = (1 - \gamma)\pi^* + \gamma\tilde{\pi}. \quad (1)$$

²This view that the beta is affected by the form of regulation is somewhat confirmed in the cross-sectional empirical evidence of [Alexander et al. \(1996\)](#).

The target profit π^* is guaranteed; therefore, Eq. (1) says that the level of profit with profit sharing can be expressed as a linear combination of the risk-free target profit and the risky price-cap profit.

For the purpose of the paper, Eq. (1) is the central relation required for our empirical analysis. The equation implies that regardless of the factors that determine return, the company's exposure to each factor with profit sharing should be γ of the exposure in the absence of profit sharing. However, to provide further insight it is useful to explore an example of how this relation could theoretically arise in the single-factor case. The mechanism in this example is straightforward. An increase in the market index leads to an increase in wealth and permanent income, which in turn causes an increase in consumption and hence profitability for the regulated company.³ Hence, there is a positive correlation between the company's value and the market index, a correlation that will depend on the form of price control regulation that a company faces. We assume that the aggregate demand for the company's output depends on aggregate wealth (this is equivalent to assuming that the consumers' utility functions satisfy "Gorman's polar form"). If firm i faces price-cap regulation, its profit, $\tilde{\pi}$, is given by

$$\tilde{\pi} = p_i x(\underline{p}, W) - c(w, x(\underline{p}, W)),$$

where x is aggregate demand for product i , c is the firm's cost function, p_i is its output price, $\underline{p}$ is the vector of prices in the economy, w is all other input prices, and W is aggregate wealth. The Gorman polar form implies aggregate demand of the form

$$x(\underline{p}, W) = a(\underline{p}) + b(\underline{p})W,$$

where a and b are functions of the vector of prices in the economy. If total costs take the form of a fixed cost, F , and marginal cost, c , then it follows that

$$\tilde{\pi} = (p_i - c)(a(\underline{p}) + b(\underline{p})W) - F.$$

The systematic risk of the regulated company within the CAPM framework can be expressed as

$$\beta_i = \frac{\text{cov}(\tilde{\pi}, W)}{\sigma_W^2} = \frac{\sigma_{\tilde{\pi}}}{\sigma_W} = (p_i - c)b(\underline{p}).$$

Then, since $\text{cov}(\pi^*, W) = 0$, the beta of a firm under the profit sharing regulation, $\hat{\beta}$, is

$$\hat{\beta}_i = \frac{\text{cov}(\tilde{\pi}, W)}{\sigma_W^2} = \gamma(p_i - c)b(\underline{p}) = \gamma\beta_i. \quad (2)$$

In other words, the profit sharing beta is the fraction γ of the price-cap beta. The more profit sharing, i.e., the lower the value of γ , the larger the difference in betas between profit sharing and price-cap regulation. Therefore, this example predicts that, other things being equal, a shift to profit sharing should reduce the risk of the regulated company to a fraction $\gamma (< 1)$ of the risk in the absence of the shift to profit sharing.⁴

³Note, [Ostergaard et al. \(2002\)](#) show that deviations from permanent income hypothesis consumption are small once one moves to state-level data, and [Attanasio et al. \(2002\)](#) show that for shareholders consumption is particularly sensitive to wealth.

⁴Following [Rubinstein \(1973\)](#), a similar process extends to other co-moments between R_i and R_m . This is a special case of the general point made earlier that given Eq. (1) and the fact that π^* is a fixed target, under profit sharing the company's exposure to any risk factor is γ of what it would be under price-cap regulation.

3. Data

3.1. Background

As mentioned in the introduction, event studies of significant regulatory changes face particular problems. Binder (1985) identifies three common problems. First, major regulatory events usually involve no single unique announcement date. Second, regulation changes often favor some firms but harm others (e.g., protect small firms at the expense of large). Finally, regulation usually affects firms in the same industry at the same calendar time and thus are difficult to separate from industry-specific shocks. In this study we are concerned with a proposed change to profit sharing for all regulated companies in the U.K. (covering airports, electricity, gas, telecommunications, and water), so the study is not exposed to the last problem at the industry level, although we do have to construct a control sample to fully interpret the results. With regard to the second problem above, the proposed change was not selectively applied. For this reason, all regulated companies in our sample are affected by the change and, more importantly, the direction of the theoretical prediction on risk is the same for all companies. Of course, the scale of the impact may differ among companies and thus, in addition to looking at the overall effect on a regulated sample, we adopt a company-by-company investigation to allow for differential effects.

The first problem, however, is a real concern for all major regulatory changes. While in some cases it is possible using daily data to identify some effects of individual regulatory announcements and publications, for a major regulatory change there will be many such dates, and it is usually the net effect of the regulation that is sought, not changes in expectations along the way. That said, the announcement problem is less of an issue for a study such as this that seeks to address the impact on risk than for studies that seek to identify transfers of wealth that arise from a major regulatory change. This is because the former is not concerned with specific share price changes on particular dates. Instead, we are concerned about whether the time-varying risk for the relevant companies was lower for a significant period of time coinciding with the time that the regulatory change was expected.

3.2. The relevant dates

Price-cap regulation has been used in the U.K. as the basis of regulation for all the utilities and, save for minor changes, the price-cap regime remained similar in each regulated sector to that introduced at privatization. Prior to their election, it was widely expected that the New Labour Party would implement an overall utilities review. Indeed, after coming to power in 1997 this occurred. The formal opening consultation for the review was published on June 30, 1997. The consultation required the sector-specific regulators to submit their views on potential changes to the current regulatory structure and specifically to indicate whether profit sharing should be adopted. Therefore, we believe that from mid-1997 onwards it is reasonable to assume that the market expected profit sharing to be a real possibility. Specifically, we take July 1, 1997, as the opening period for the expected change.

The regulators expressed reservations about a shift to profit sharing. Nonetheless, it was clear that the government was not going to lay profit sharing to the side. The March 1998

formal Green Paper (Department of Trade and Industry, 1998a) explicitly favored the use of error correction mechanisms as “an in-built means of sharing unearned benefits promptly between consumers and shareholders”. As always with statements of this type, there were prior discussions between interested parties and the government as the ideas became concrete. Therefore, it is difficult to pinpoint the exact date upon which information showing that the government was fully committed to profit sharing entered the market. By January 1998 this information was being reported widely in the newspapers, e.g., the *Financial Times* published an article, ‘Utilities brace for wind of change’, indicating that discussions between the parties were not going as well for the companies as they would have liked.⁵ Therefore, the strength of the government’s intentions would have been known many months before January 1998. By February much of the debate focused on the practical details. For example, the *Independent* pointed out that “it is not clear how much detail the Green Paper will contain on what constitutes ‘excess’ profit, how it would be taxed and what shareholders might be allowed to retain”.⁶ On March 1st the *Financial Times* confirmed the view that profit sharing would be introduced.⁷

The regulated companies were against profit sharing and lobbied to get the proposals overturned. Apparently the issue became a battle within the government, with some early supporters pulling back from the proposals because of the “poor” signal it gave to the business community of the intentions of the government. In July 1998, the front page of the *Financial Times* claimed that the Prime Minister himself might intervene to prevent the government minister in charge of the Department of Trade and Industry from implementing the proposals. The article claimed that Tony Blair was “set to veto proposals to force utilities to share windfall profits with consumers through changing the system of price regulation”, and if correct “the decision to scrap the long-standing Labour plan will be welcomed by the city”.⁸ Later that month the *Financial Times*, referring to utility regulation, still claimed that “the most controversial issue has been the long-standing Labour proposal to add some form of profit sharing to the existing inflation-minus-X price regulation formula”.⁹

Despite the pressure from the Prime Minister, when the proposals were brought forward again (Green Paper, Department of Trade and Industry, 1998b) the government still had not dropped the idea of profit sharing. It did, however, dilute the proposal to some extent. Specifically, the second Green Paper stated that it might be appropriate to refine the price formula when companies made profit by good fortune, and “in such cases regulators would be expected to pass on to customers company profits made thanks to specific factors outside their control”.¹⁰ After this second Green Paper the minister in charge of the Department of Trade and Industry was replaced with one more favorable to business and over time the debate on profit sharing gradually dissipated. Formally, the relevant Utilities Bill entered Parliament for discussion at the start of 2000. The Utilities Bill received “Royal Assent” in July 2000 and contained no profit sharing regulations. One could take this as the closing date of our treatment period, but this would be misleading. It was clear

⁵January 23, 1998.

⁶‘Utilities braced as Government plans new tax on excess profits’, *Independent*, February 23, 1998.

⁷‘Beckett to propose profit cap in utilities shake-up’, March 1, 1998.

⁸‘Blair set to veto proposals on utilities’ windfall profits’, *Financial Times*, July 10, 1998.

⁹‘Utility chiefs face pay and service level link’, *Financial Times*, July 27, 1998.

¹⁰*Financial Times*, July 28, 1998.

from the parliamentary debate before the Bill's "Royal Assent" that there would be no profit sharing of any type. Indeed, we argue that even before early 2000 it was obvious that profit sharing was no longer a genuine possibility.

We suggest that a critical date is that of August 12, 1999. On this date the detailed draft proposals for all the Regional Electricity Companies were simultaneously published (for final consultation). These proposals had no profit sharing element in them at all and it has always been the case that, although the proposed prices may have changed from final proposal to implementation, the structure of the regulation did not. That is, all the evidence on that date pointed to there being no profit sharing for any of the electricity companies. Therefore, we believe, it is very difficult to argue that profit sharing was still a likely outcome in the U.K. after this. For this reason we close the period of policy uncertainty with this date. Indeed, as we see below, the opening and closing dates for the profit sharing period of July 1, 1997 and August 12, 1999 are confirmed by our results obtained using the Kalman filter approach.

The profit sharing period is only part of a longer sample period taken for the investigation. Specifically, we employ daily and monthly data on the companies from May 1, 1993 to December 29, 2000. All the time series are collected from DataStream and are dividend adjusted. In the case of monthly observations we use end-of-month log-returns. This gives 1,938 log-returns at the daily frequency for most companies. The starting and closing dates are chosen to balance the desire to include a long period of price-cap regulation (i.e., either side of the profit sharing period) against the constraint of analyzing as many regulated companies as possible (mergers and breakups reduce the number of companies the longer the sample period adopted), while also avoiding having a large number of companies in the sample that are in their immediate post-privatization position. The construction of our samples is described in detail in Sections 3.3 and 3.4.

3.3. *The sample of regulated companies*

The privatization program initiated by the Conservative government introduced over 30 public utility companies to the London Stock Exchange in the 1980s and early 1990s. However, many of these utilities have either been acquired or merged since privatization. For example, of the original 12 regional electricity companies (RECs), none remain listed on the stock exchange in 2000. As a result, only 22 companies are listed on the London Stock Exchange between May 1993 and December 2000. Among these 22 companies three were listed for a slightly shorter period of time (Viridian listed since June 18, 1993, National Grid Company listed since December 8, 1995 and Hyder delisted in October 2000) but are included to maintain sample size. Extending the overall sample period beyond December 2000 would further reduce the sample size. Unfortunately, we had to delete seven companies from our analysis due to poor data quality: six extremely small water companies suffer from thin trading, making the data unsuitable for any statistical and time-series analysis, and British Gas where there was a fundamental restructuring and separation.

The remaining 15 companies are controlled by price-cap regulation with the exception of PowerGen and National Power. These two are regulated but do not face direct price controls. We keep PowerGen and National Power in the sample as the profit sharing regime could have been expected to apply to them directly if the Utility Review had increased the scope of regulation, in which case it would not be unreasonable to think that

their potential profitability could have been similarly affected, albeit indirectly, by these changes. Note that all companies in the sample, save BT and to a lesser extent the National Grid Company, can be described as “old economy”. BT is a clear outlier and we give special attention to it and National Grid Company when analyzing our findings.¹¹ Appendix A lists the 15 regulated companies included in the sample.

3.4. *The control sample*

A potential problem in isolating the specific effects of the regulatory events is that there may be other changes that could have affected the performance of regulated companies relative to the market. A particular concern in this regard is that the composition of the market changed towards the end of our sample period because e-business and related stocks became extremely significant. A particular feature of these stocks was that there was considerable uncertainty as to their long-run position. That is, they were more like options on underlying assets than is the case for conventional shares. While this feature helped push up the returns on the stock market portfolio, it also could have increased the risk of holding that portfolio. For many companies not in the high-tech sector, this may have manifested itself as a decrease in their betas. Since most regulated companies are traditional utilities they are likely to be old economy-type stocks and prone to any such effect. For this reason it is useful to create a control sample to attempt to eliminate any such effect.

A standard approach to create a control is to match individual firms with similar firms within the same sector. Clearly this option is not available here, since we attempt to isolate effects that are specific to regulated sectors. Thus, the control sample has to be drawn across other sectors. There are various properties that we desire in the construction of the control sample. For instance, while the control sample should not contain any regulated companies, it should otherwise attempt to be representative of the general market save for the elimination of companies that may strongly benefit from any e-business effect. However, in addition to being representative of the market, it would also be useful if the control sample were similar to the regulated sample in terms of number of firms, size of firms, and leverage ratios. Inevitably there has to be some compromise between these demands.

In the U.K. there was, until 2003, a 30-company index (FT30 index) that consisted of 30 companies that were chosen to be representative, as a group, of the U.K. economy. Clearly, identifying such a small number of companies as representative of the market requires an element of judgment. The FT30 was chosen by a committee and was sometimes criticized as an index because it consisted of the view of the committee as to what they deemed to be representative rather than being objective, say, in the sense of the FTSE100, which is basically composed of the largest one hundred companies on the market. However, this representative albeit subjective feature is exactly what we require of our control sample. We therefore use as our starting point for the construction of our control portfolio the 30 FT30 companies at the end of our sample period.

¹¹National Grid Company had placed a small telecommunications network around its electricity network and created a telecommunications company called Energis (within the quoted National Grid Company business). While National Grid Company is in part “old” economy and, in part, might benefit from the growth in e-commerce, given the relative size of the two components, it tends to be closer to the former.

We remove from this list all regulated companies and those that are likely to be directly affected by the new economy, namely, companies in financial, electronic, information, communications, and media sectors. Appendix A gives the list of companies and those that are eliminated. This leaves 18 companies in the control sample (compared to 15 in the regulated sample). The average capitalization at the start of our sample period was £4,005m for the regulated sample and £5,507m for the control sample. The regulated sample contains one company (BT) that is far larger than the others in the regulated sample, and the control sample has two large outliers (British Petroleum and Glaxo Wellcome). The average leverage for the first full year of the sample period was 0.10 for the regulated sample and 0.11 for the control sample.

4. Empirical analysis: Market model

4.1. Model selection and specification

Although addressed in many research papers, the best way to model an asset's exposure to risk remains unresolved. Historically the CAPM approach was dominant but in recent years the three-factor model developed by Fama and French (FF3F) (Fama and French, 1992, 1993) is frequently used. In this section we analyze the time-varying beta in a single-factor market model and in Section 6 we consider the extent to which the FF3F model provides further insight.

The time-varying nature of CAPM betas has been reported for more than 30 years but only in the last 15 years has there been significant progress in the development of econometric techniques that enable the estimation of the time pattern of market risk. There are now many such papers, e.g., Bollerslev and Zhang (2003), Braun et al. (1995), Schwert and Seguin (1990), and Lettau and Ludvigson (2001). A particular issue is that the violation of assumptions of normality, identity, and independence in stock market returns focuses attention on the most suitable estimation technique and appropriate specification of the error generating process (in particular, the pattern of changes in volatility). We address the problem by enriching the CAPM model with a GARCH(1,1) specification of the variance term and employing the Kalman filter technique to estimate the GARCH effects and the time paths of CAPM coefficients. We also test for a stochastic time trend but do not report the results since no such trend is detected. The results can be obtained from the authors on request.

For the purposes of this paper the use of the Kalman filter is more suitable than the switching-regression model. This is primary because, if we use the Kalman filter, we are not obliged to define the exact point of the switch. Instead, whether and when there is a change in the pattern of beta can be assessed after the regression results are revealed. This is particularly important when, as in our case, the timing of the change is well understood but is not unambiguously defined. A second benefit of our modeling choice is that switching models are at best numerically demanding and in the case of short data periods are impossible to implement.¹²

¹²This is a particular problem for the FF3F analysis in Section 6 since the last two subperiods, i.e., the profit sharing period and post-profit sharing period, account for 25 and 17 monthly observations, respectively. Bearing in mind that even a Markov switching model with just a mean and a variance would require the definition of four states, and a transition matrix containing 12 independent elements, the Kalman filter appears the best solution.

We use the market model specification, i.e., we assume that, at any time t , the return on asset i , $R_{i,t}$, can be explained by the return on the market portfolio $R_{M,t}$:

$$R_{i,t} = \alpha_{i,t} + \beta_{i,t}R_{M,t} + \varepsilon_{i,t}, \quad (3)$$

where the error generating process $\varepsilon_{i,t} \sim N(0, \sigma_{i,t}^2)$ has a variance described by a GARCH(1,1) process.¹³ The time-varying coefficients $\alpha_{i,t}$ and $\beta_{i,t}$ constitute a state vector in the state-space representation of the Kalman filter. Denoting the value of the state vector at time t by $a_{i,t}$ (i.e., $a_{i,t} = [\alpha_{i,t}, \beta_{i,t}]'$), we complete the state-space model by defining the state equation as

$$a_{i,t} = a_{i,t-1} + \eta_{i,t}, \quad (4)$$

where $\eta_{i,t}$ is a 2×1 vector of serially uncorrelated disturbances with zero mean and covariance matrix Q . In fact, to express the above system in the state-space representation of the Kalman Filter, the state vector should have three elements: the two already-mentioned stochastic parameters of the measurement equation and a deterministic element to represent the variance of the error term, σ_i^2 . To conduct the calculations we adopt the technique of parameter estimation introduced in Zalewska-Mitura and Hall (1999). The random walk specification implies a filter on the data in which parameters evolve smoothly and are mostly determined by the observations around time t . How much data around time t is used for estimation of the coefficients, (i.e., α_t and β_t) depends on the variance of η_i and is estimated from the data. The specification is therefore well suited for depicting the likely evolution of market risk.

The above state-space specification requires the estimation of six unknown parameters before the time paths of α_t and β_t can be calculated using the Kalman filter algorithm. These six parameters are: the two diagonal elements of matrix Q , the three GARCH specification parameters, and the initial value of the variance of the disturbance term in the GARCH specification. However, it is more convenient for the initial value of the disturbance term to be concentrated out of the likelihood function and determined after the values of the remaining five unknown parameters are obtained from maximization of the corresponding concentrated likelihood function. Finally, a fixed-interval smoothing algorithm is applied.

In addition to the above specification we also use, but do not present, the CAPM specification as our measurement equation, i.e.,

$$R_{i,t} - r_{f,t} = \beta_{i,t}(R_{M,t} - r_{f,t}) + \varepsilon_{i,t},$$

where the state vector is defined as $a_{i,t} = [\beta_{i,t}]$ and its corresponding disturbance term $\eta_{i,t}$ is normally distributed with zero mean and variance q . Although the latter representation has the slight advantage of reducing the number of estimated parameters to five (we do not need to estimate the variance for the equation of the constant term $\alpha_{i,t}$), a choice has to be made concerning the proxy for the risk-free rate of return. In our analysis we use both ten-year and 15-year T-bonds as proxies for the risk-free rate.¹⁴

¹³GARCH coefficients are not presented but can be obtained from the authors upon request. Almost all coefficients are significantly different from zero at the 1% level and strongly indicate the presence of heteroskedasticity.

¹⁴Our decision to use long-term T-bonds, rather than T-bills, as a risk-free rate of return is based on the notion that the former may better match companies' investment horizons.

The estimated time paths of the state vector components are virtually identical whether ten- or 15-year T-bonds are used. They are also twin-similar to the time paths estimated for the market model representation specified in Eq. (3). The only difference between the CAPM and market model estimates is in the GARCH effects, and this difference is small. Since the main results are the same for all time-series specifications we present only the market model here (as specified by Eq. (3)). Furthermore, since the time paths estimated for the constant term are both virtually zero and insignificantly different from zero in all regressions, we report time paths of the beta coefficient only.

As explained in Section 3 we have a sample of 15 regulated companies and a control sample of 18 companies. To test for the difference in the behavior between these two groups we create two equally weighted portfolios, one for the regulated sample and one for the control sample. Note that because the CAPM beta is additive, estimating betas for a portfolio is equivalent to estimating our model for each company separately and then averaging the betas. However, the portfolio estimate has an advantage over the average of the individual estimates in that it automatically provides confidence intervals for the estimated time paths. Therefore, we perform our analysis on an equally weighted portfolio of regulated companies and an equally weighted portfolio of control sample companies as dependent variables. In the remainder part of the paper, to simplify notation we refer to the time paths estimated for the portfolio of the regulated companies as the *regulated beta* and to the time paths estimated for the portfolio of the control sample companies as the *control beta*. The market portfolio, R_M , is represented by the log-returns on the FTSE All Share index.

We use daily data in this section and monthly data in Section 6. Our decision to limit the analysis in Section 6 to monthly data is dictated by the frequency at which additional factors for the FF3F model are available.

Finally, it is important to explain how we test for the significance of our results. As we explain in Sections 2 and 3, we expect differences in risk between our two portfolios during the profit sharing period, i.e., between July 1, 1997 and August 12, 1999. Although the fundamental properties of our portfolios may be different between the pre-profit sharing period (May 1, 1993–June 30, 1997) and the post-profit sharing period (August 13, 1999–December 31, 2000), these differences should be similar for the regulated and the control portfolios. In other words, if profit sharing is a central driving force of the differences between samples, then differences in risk, if any, between the portfolios should be constant when this factor is absent, regardless of whether the level of risk itself changed. More precisely, if $R_{\text{reg},t}$ denotes the return on the regulated portfolio, $R_{c,t}$ is the return on the control portfolio at time t , and it is true that both

$$R_{\text{reg},t} = \alpha_{\text{reg},t} + \beta_{\text{reg},t} R_{M,t} + \varepsilon_{\text{reg},t} \quad (5)$$

and

$$R_{c,t} = \alpha_{c,t} + \beta_{c,t} R_{M,t} + \varepsilon_{c,t}, \quad (6)$$

then the difference between the return on the regulated portfolio and the return on the control portfolio can be explained as a function of the market portfolio return:

$$R_{\text{reg},t} - R_{c,t} = \alpha_{\text{diff},t} + \beta_{\text{diff},t} R_{M,t} + \varepsilon_{\text{diff},t}, \quad (7)$$

where $\alpha_{\text{diff},t} = \alpha_{\text{reg},t} - \alpha_{c,t}$, $\beta_{\text{diff},t} = \beta_{\text{reg},t} - \beta_{c,t}$, and $\varepsilon_{\text{diff},t} = \varepsilon_{\text{reg},t} - \varepsilon_{c,t}$. We estimate Eq. (7) using the state-space representation as introduced in Eqs. (3) and (4). The difference in the

betas should be greater during the profit sharing period than in the other periods. It is important to note that a random walk specification of Eq. (4) for both the α and β coefficients is still valid for the difference equation, Eq. (7).

4.2. Kalman filter estimation results

As described in Section 3.2 there are three distinctive periods in our sample. To show differences in risk between the regulated and control portfolios we separately estimate beta time paths for the portfolios for each of the subperiods. The three disjoint subperiods are as follows:

- (1) Pre-profit sharing period: May 1, 1993–June 30, 1997. If our hypothesis is true, both the regulated and control portfolios should exhibit a similar pattern (save that the regulated betas may be slightly lower than the control betas).
- (2) Profit sharing period: July 1, 1997–August 12, 1999. The time path estimated for the regulated portfolio in this period should be significantly below the time path estimated for the regulated portfolio in the pre-profit sharing period. This should not be the case for the control portfolio.
- (3) Post-profit sharing period: August 13, 1999–December 31, 2000. The relative risk for the regulated and control portfolio should be similar to that observed in the pre-profit sharing period. However, because this period covers the period of the e-business effect, we conjecture that the absolute values may be below the values estimated for the first subperiod.

Fig. 1 presents time paths of the betas estimated for the portfolios of the regulated (thick line) and control (thinner line) portfolios for each of the subperiods separately. We plot the results of the six regressions (i.e., three subperiods for each of the two portfolios) on one graph. The pre- and post-profit sharing periods exhibit considerable similarity between the regulated and control portfolios.

Considering the control portfolio first, it is clear that, although the time paths for the control betas come from three independent regressions, there is almost no discontinuity when moving between subperiods. The plots look almost as if the time path obtained came from a single regression using all the data points for the period 1993–2000 at once. That is, although none of the information from the first and third subperiods is used to derive the beta of the middle period, and similarly, none of the information of the middle period is used to derive the betas of the first and third subperiods, there is almost no difference between the end of each time period and the start of the next. This suggests that there is no change in the underlying process that generates the data of the control portfolio around the critical dates.

However, the situation is very different when we come to the regulated portfolio. The profit sharing betas display substantial diversity, with a large discontinuity in the regulated beta between subperiods. This indicates that there is a substantial change in the underlying process that generates the data of the regulated portfolio during the profit sharing period.

As we explain in Section 4.2, to have a complete comparison between the two portfolios across time, we estimate time paths of the constant term and the beta coefficient for the difference of returns between the regulated portfolio and the control portfolio (i.e., Eq. (7)). The $\alpha_{\text{diff},t}$ coefficient is virtually zero and shows almost no variability. Specifically,

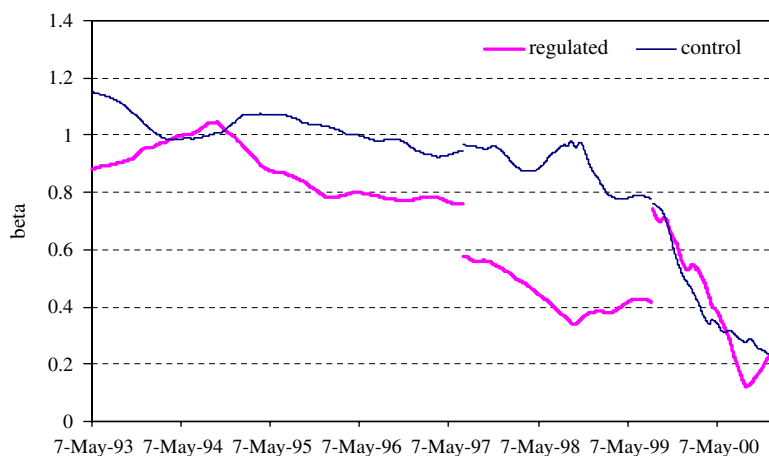


Fig. 1. Kalman filter estimates of the time paths of the betas estimated separately for the portfolios of the regulated and control samples for each of the pre-profit sharing (May 1, 1993–June 30, 1997), profit sharing (July 1, 1997–August 12, 1999) and post-profit sharing (August 13, 1999–December 31, 2000) subperiods using daily returns and the CAPM Market model specification. If there is a substantial change in the underlying process that generates data in a portfolio during the profit sharing period, there should be a large discontinuity in the beta between subperiods.

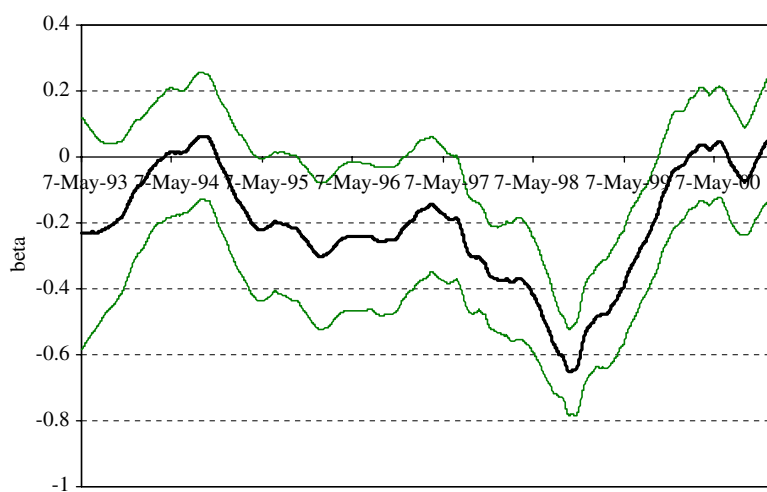


Fig. 2. Kalman filter estimates of the CAPM Market model beta coefficient for the difference in returns between the regulated and control portfolios over the period May 1, 1993–December 31, 2000 using daily observations. The thick line shows the coefficient and the two thin lines show the 95% confidence intervals.

its values vary between 0.0000609 and 0.0000619, and stay within the 95% confidence intervals of approximately $(-0.00029, 0.00042)$. However, the time path of the $\beta_{\text{diff},t}$ coefficient shows a very different pattern.

Fig. 2 presents both the time path of the $\beta_{\text{diff},t}$ coefficient (thick line) and 95% confidence intervals (thinner lines). It is clear that during the profit sharing period the difference in

market risk between the two portfolios in question is significantly different from zero. In contrast, the $\beta_{\text{diff},t}$ coefficient in both the period before and after the regulatory debate is not statistically different from zero. This result is confirmed when we estimate coefficients of Eq. (7) for each of the three subperiods independently. We also repeat the whole exercise conducted in Section 4.2 using weekly data and find no differences in results.

We conduct the above results on the regulated and control portfolios. However, we also wish to observe the phenomena for individual companies, both to ensure that it is present at the disaggregated level and to identify any companies that may exhibit different trends. The latter point is important since there are some differences across companies in the regulated sample. In particular, telecommunication companies (i.e., BT and to a lesser extent National Grid Company) may exhibit a different pattern over time, since they could be affected differently by the e-commerce bubble than the remaining companies in the sample.

The company-level analysis uses Eq. (7) as the measurement equation, where $R_{\text{reg},t}$ represents the return at time t for an individual regulated company rather than the return on the equally weighted portfolio of all the regulated companies included in the study. The comparator, $R_{c,t}$, continues to be the equally weighted average return for the control sample at time t . The estimated time paths can be found in Appendix B. We also estimate subperiod time paths for all 15 regulated companies. The subperiod results are very similar to, and even stronger than, the results obtained from regressions for the whole period, i.e., 1993–2000. For brevity, we present the regressions for the whole period only. However, the results for the subperiods can be obtained from the authors upon request.

There is a strong pattern (i.e., larger difference in systematic risk between the company and the control portfolios during the profit sharing period) for all companies but one in the sample. That is, the results obtained with the regulated portfolio are confirmed at the company-level analysis. The outlier is the National Grid Company. However, the National Grid Company “anomaly” could have arisen because the pre-profit sharing period estimates are based on a period during which the National Grid Company was newly listed. Therefore, the National Grid Company results may be driven by strong company-specific factors. Both BT and the National Grid Company exhibit a relative increase in their systematic risk in the post-profit sharing period. This is consistent with the fact that both companies (BT as a whole, and the National Grid Company partly) are telecommunication services providers and hence would be affected differently by the e-commerce bubble than other regulated companies. Finally, BAA is the only company where the support for a decrease in beta during the profit sharing period is weak. However, even BAA has lower values of the beta coefficient in the profit sharing period, but this difference is not significant.

Finally, to complete this section it should be mentioned that the results remain unaffected when the control sample is divided into subgroups. To create divisions that generate the maximum difference (i.e., four companies shift groups), we first cut the control sample “randomly” by taking the first and then the latter half of the sample when ranked alphabetically. We then repeat the process by cutting the control sample into the largest and smallest companies according to December 2000 market capitalization.¹⁵ The divisions do not affect our results, i.e., all the differences between the regulated portfolio

¹⁵We do not provide the results of regressions on the difference between the regulated sample and each individual company in the control sample since this would require 18 graphs.

and the subsamples of the control portfolio are significant at the 5% level within the profit sharing period (see Appendix C for figures equivalent to Fig. 2).

4.3. OLS WHC and ML with GARCH(1,1) estimation results

To complete our discussion of the differences in systematic risk between the regulated and the control portfolios, and to obtain some feel for the significance of these effects, we also adopt a more traditional approach and assume that, save for the specific changes investigated here, the betas are fairly constant over subperiods (indeed, this is true to different degrees for the above-defined subperiods). In this case it is possible to introduce a profit sharing dummy, D_{ps} , where D_{ps} takes the value of one between July 1, 1997 and August 12, 1999, and zero otherwise. We conduct the analysis on the returns of the regulated portfolio, on the difference between the returns of the regulated and control portfolios, and on the differences between the returns for individual regulated companies and the returns on the control portfolio. More precisely, the regression takes the form

$$R_{i,t} = \alpha + \alpha_{ps}D_{ps} + \beta R_{M,t} + \beta_{ps}D_{ps}R_{M,t} + \varepsilon_t,$$

where $R_{i,t}$ refers either to (i) the return on the regulated portfolio, $R_{reg,t}$; (ii) the difference between returns on the regulated portfolio and control portfolio, $R_{reg,t} - R_{c,t}$; or (iii) the difference between returns on an individual company ($R_{I,t}$) and the return on the control portfolio, i.e., $R_{I,t} - R_{c,t}$. Estimates are obtained using OLS WHC and ML GARCH(1,1). We use OLS to obtain a consistent comparison of the monthly and daily results (ML analysis is not possible for the lower frequency data due to the small sample size). However, ML GARCH(1,1) provides a comparison of results of the time-varying and time-invariant models applied to daily data.

Table 1 presents the OLS WHC and ML GARCH(1,1) results when the difference between the returns on the regulated and control portfolios, and the returns on the regulated portfolio, are used as a dependent variable, and it presents OLS WHC and ML GARCH(1,1) results using the difference between the returns of individual companies and the returns of the control portfolio. For all the regressions but two individual companies the profit sharing dummy, β_{ps} , is significant at the 1% level and has the correct sign. This is strong confirmation that the regulatory change had the predicted effect on market risk.

If the market model is the correct empirical specification then profit sharing should have an effect on β and not α . On the other hand, if the market model is a proxy for the CAPM, then theoretically there should be an effect on α , but the values of α_{ps} and α should be miniscule relative to those of β_{ps} and β . Either way, any effect we observe on α_{ps} and α should be minute relative to any effect on β . This is exactly what we find in Table 1.

We do not present adjusted- R^2 statistics. Most regressions are calculated either on the differences between the returns on two portfolios, or on the difference between company returns and the returns of the control portfolio. Since the difference in the betas of the two portfolios (save for the profit sharing period) is assumed to be constant regardless of the market portfolio return, the R^2 s are likely to be low. Indeed, the R^2 for most regressions using differences is between 5% and 10% as expected. In the case of regressions that use the returns on the regulated portfolio as a dependent variable, the adjusted- R^2 is 26%.

Table 1

Regression results of the model $R_{i,t} = \alpha + \alpha_{ps}D_{ps} + \beta R_{M,t} + \beta_{ps}D_{ps}R_{M,t} + \varepsilon_t$.

At the “Aggregate level” $R_{i,t}$ is represented either by the difference between returns on the regulated portfolio and control portfolio ($R_{reg,t} - R_{c,t}$), or returns on the regulated portfolio ($R_{reg,t}$). In the “Individual company level” panel the dependent variables are defined as the difference between returns on individual company (indicated by its name) and the return on the control portfolio. Regressions are based on daily observations covering the May 1, 1993–December 31, 2000 period. Dummy D_{ps} is equal to one between July 1, 1997 and August 12, 1999, and zero otherwise.

	OLS WHC				ML GARCH(1,1)			
	α	α_{ps}	β	β_{ps}	α	α_{ps}	β	β_{ps}
<i>Aggregate level</i>								
$R_{reg,t} - R_{c,t}$	−0.000 (0.000)	0.000 (0.000)	−0.056 (0.037)	−0.416*** (0.055)	0.000 (0.000)	0.001 (0.000)	−0.112*** (0.028)	−0.325*** (0.037)
$R_{reg,t}$	−0.000 (0.000)	0.000 (0.000)	0.638*** (0.038)	−0.208*** (0.049)	0.000 (0.000)	0.000 (0.000)	0.727*** (0.030)	−0.268*** (0.043)
<i>Individual company level</i>								
Anglian Water	−0.000 (0.000)	0.000 (0.001)	−0.165** (0.077)	−0.550*** (0.103)	0.000 (0.000)	0.001 (0.001)	−0.238*** (0.054)	−0.386*** (0.070)
BAA	−0.000 (0.000)	0.000 (0.001)	−0.114* (0.067)	−0.084 (0.091)	0.000 (0.000)	0.000 (0.001)	−0.029 (0.042)	−0.148** (0.067)
BT	−0.001* (0.000)	0.002* (0.001)	0.677*** (0.101)	−0.437*** (0.132)	−0.001* (0.000)	0.002*** (0.001)	0.233*** (0.049)	−0.075 (0.085)
Hyder	−0.000 (0.001)	−0.000 (0.001)	−0.248** (0.111)	−0.520*** (0.123)	0.000 (0.000)	0.000 (0.001)	−0.465*** (0.047)	−0.286*** (0.060)
Kelda	0.000 (0.001)	−0.001 (0.001)	−0.251** (0.101)	−0.516*** (0.125)	0.000 (0.000)	0.000 (0.001)	−0.323*** (0.058)	−0.385*** (0.077)
National Grid Company	0.000 (0.001)	0.001 (0.001)	−0.223** (0.105)	−0.484*** (0.132)	0.000 (0.000)	0.001 (0.001)	0.084 (0.070)	−0.340*** (0.085)
National Power	−0.000 (0.001)	−0.001 (0.001)	0.054 (0.129)	−0.410*** (0.159)	0.000 (0.000)	0.000 (0.001)	0.042 (0.052)	−0.342*** (0.080)
Pennon	−0.000 (0.000)	0.001 (0.001)	−0.400*** (0.090)	−0.369*** (0.102)	−0.001*** (0.000)	0.001*** (0.000)	−0.639*** (0.031)	−0.052 (0.049)
PowerGen	0.000 (0.000)	−0.001 (0.001)	−0.031 (0.080)	−0.413*** (0.113)	0.000 (0.000)	−0.001 (0.001)	−0.146*** (0.052)	−0.227*** (0.070)
Scottish Power	−0.000 (0.000)	−0.001 (0.001)	0.116* (0.067)	−0.440*** (0.099)	0.000 (0.000)	0.001 (0.001)	0.008 (0.054)	−0.334*** (0.084)
Scottish & Southern Energy	0.000 (0.000)	0.001 (0.001)	−0.060 (0.075)	−0.553*** (0.104)	0.000 (0.000)	0.001 (0.001)	−0.219*** (0.051)	−0.417*** (0.067)
Severn Trent	−0.000 (0.000)	0.000 (0.001)	−0.143** (0.073)	−0.434*** (0.100)	0.000 (0.000)	0.000 (0.001)	−0.146*** (0.057)	−0.393*** (0.078)
Thames Water	0.000 (0.000)	0.000 (0.001)	−0.195** (0.078)	−0.197* (0.110)	0.000 (0.000)	0.001 (0.001)	−0.186*** (0.048)	−0.144** (0.058)
United Utilities	−0.000 (0.000)	0.000 (0.001)	0.090 (0.068)	−0.498*** (0.103)	0.000 (0.000)	0.000 (0.001)	0.024 (0.058)	−0.455*** (0.075)
Viridian	0.000 (0.000)	0.000 (0.001)	−0.318*** (0.058)	−0.407*** (0.082)	0.000 (0.000)	0.001 (0.001)	−0.360*** (0.037)	−0.365*** (0.060)

OLS WHC and ML GARCH(1,1) indicate that the estimates were obtained by OLS with White heteroskedasticity consistent standard errors and covariance corrections, and with ML estimations with a GARCH(1,1) error specification, respectively.

Standard errors are in parentheses. *, ** and *** denote significance at the 10%, 5%, and 1% level, respectively.

5. International comparison and leverage effects

The effects reported in Sections 4.2 and 4.3 are strong. Although the timing of the decline in market risk is aligned with the profit sharing debate, it is important to show that it cannot be attributed to other factors such as a change in worldwide conditions or a change in the debt structure of the analyzed companies. To address this issue we provide a comparison of market risk time paths for comparable portfolios in the U.K. and U.S. (Section 5.1) and show that because the decline in market risk during the profit sharing period is not observable in the U.S. market, it is therefore not caused by a worldwide phenomenon. Further, in Section 5.2 we discuss debt structure changes of our sample companies and show that these changes cannot explain the different market risk time paths for the two portfolios.

5.1. U.S. and U.K. comparison

The aim of this subsection is to draw a comparison with the U.S. market and thereby to control for changes in supply and demand conditions (worldwide) that may impact on U.K. companies but that are unrelated to the regulation of ownership structure inside the U.K.

During the full sample period there are two changes that we observe in the estimated risk of the portfolios. The first is a decrease in the market risk of the regulated portfolio (but not the control portfolio), occurring between July 1997 and August 1999. This decrease is attributed to the regulatory debate, which intensified during that time. The second observed change is a decline in the estimated values of market risk, occurring in the post-profit sharing period, i.e., between late 1999 and December 2000. This change is present in both the regulated and control portfolios. We conjecture that this is a result of the e-commerce bubble. If our interpretation of the observed U.K. phenomena is correct, then, when we replicate our analysis on comparable U.S. data, (i) we should not observe any discrepancy between equivalent U.S. portfolios during the profit sharing period, but (ii) we should observe a decline in the estimates of the betas for both groups of companies after 1999 since, while the companies in the U.S. should not be exposed to any change in the regulatory regime, they should display similar sensitivity to the e-commerce bubble.

To construct comparable U.K. and U.S. samples we use DataStream (DS) sector indexes. We approach the problem in two steps. First, we take DS sector indexes for the U.K. that match the U.K.'s regulated and control samples. Then we select the equivalent US-DS indexes. To construct the UK-DS regulated portfolio, we use the Water, Electricity, and Telecom Fixed Line DS indexes.¹⁶ The regulated portfolio also includes BAA (an airport company) so there might be an argument for the inclusion of the Airline and Airports index (which includes BAA) in the UK-DS regulated portfolio. However, the Airlines and Airports index also includes British Airways, a company that is in our control portfolio. We exclude the Airline and Airports index from the UK-DS and US-DS

¹⁶To avoid confusion, given that we have six portfolios, we use the following terminology throughout this subsection. The regulated and control portfolios defined in Section 3.3 and 3.4 are still referred to as the regulated and control portfolios (or samples). Their equivalents based on the DS sector indexes are called the UK- (or US-) DS regulated and the UK- (or US-) DS control portfolios (or samples). Note, we use the terminology UK-DS (or US-DS) regulated although not every company in these “regulated” portfolios are directly price regulated.

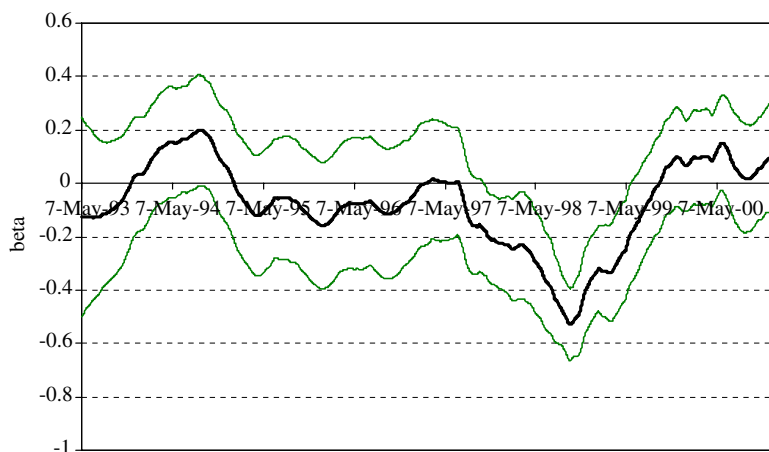


Fig. 3. Kalman filter estimates of the CAPM Market model beta coefficient for the difference in returns between the UK-DS regulated and UK-DS control portfolios over the period May 1, 1993–December 31, 2000 using daily observations. The thick line shows the coefficient and the two thin lines show the 95% confidence intervals.

regulated portfolios but include it in the UK-DS and US-DS control portfolios for two reasons. First, the U.K. Airline and the Airports index contains six other (nonregulated) companies. Second, the corresponding Airline and Airports index calculated for the U.S. market does not contain any airports. The regulated and control portfolios (based on individual companies) used throughout the paper are equally weighted. We weigh the constituent indexes in the UK-DS and US-DS regulated portfolios to reflect the distribution of companies in the regulated sample. That is, because there are seven water companies, six electricity companies, and one telecom company in the regulated portfolio (see Appendix A), the weights allocated to the corresponding UK-DS and US-DS regulated portfolios are 7/14, 6/14, and 1/14, respectively.

The UK-DS control portfolio consists of 13 sector indexes chosen to match sectors of individual control portfolio companies. Each control portfolio company belongs to one DS index only. However, the Distillers and Vintners index, Chemicals index, Food Producers and Processors index, Pharmaceuticals index, and Retailers-Departments index contain two companies each from the control sample. The 13 indexes in the UK-DS and US-DS control portfolios are equally weighted.¹⁷

Finally, for both markets, DS Total Market indexes are used as the market portfolios. The U.K. sample consists of 1,938 daily observations and the U.S. sample contains 1,939 daily observations. Fig. 3 shows the estimates of the time path of the $\beta_{\text{diff},t}$ coefficient for the UK-DS portfolios. This is the UK-DS equivalent to Fig. 2. Figs. 2 and 3 display almost identical patterns.

To show changes taking place in the U.S. market and to provide a comparison with the changes occurring in the U.K. we present two graphs for the US-DS portfolios. Fig. 4

¹⁷The regulated portfolio indexes: Water, Electricity, Telecom Fixed Lines. The control portfolio indexes: Airline and Airports, Aerospace Defence, Auto-Parts, Chemicals, Construction and Building Materials, Distillers and Vintners, Food Producers and Processors, Food and Drug Retailers, Oil Integrated, Pharmaceuticals, Retailers-Departments, Shipping and Ports, Leisure and Hotels.

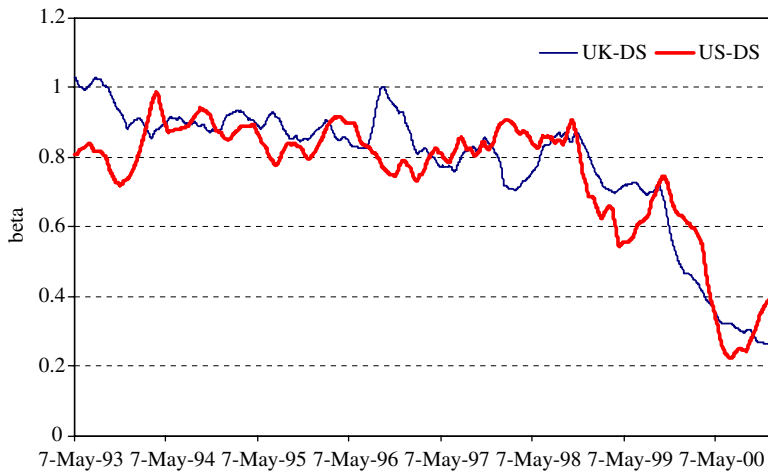


Fig. 4. Kalman filter estimates of the CAPM Market model beta coefficient for the returns on the UK-DS and US-DS control portfolios over the period May 1, 1993–December 31, 2000 using daily observations.

shows time paths of the β_t coefficient estimated for the US-DS control portfolio (thick line) and for the UK-DS control portfolio (thinner line). Because the time series of the US-DS portfolio is one data point longer than the UK-DS portfolio, for presentational purposes we delete the first estimated US-DS beta. The UK-DS and US-DS control portfolios have very similar time paths for the β_t coefficient. The similarity is present both in the level during the first two subperiods and decline during the final period (i.e., the post-profit sharing period).

Fig. 5 presents the time path of the $\beta_{\text{diff},t}$ coefficient estimated for the US-DS portfolios, i.e., the US equivalent to Fig. 3. Here, the pattern of the $\beta_{\text{diff},t}$ coefficient is very different from the one presented for the UK-DS portfolios. In particular, there is no decline in the $\beta_{\text{diff},t}$ coefficient between 1997 and 1999.¹⁸ This indicates that the observed decrease in market risk for the regulated portfolio and the UK-DS regulated portfolio is not an international phenomenon. Note that at the end of the investigated period, the difference between the betas of the US-DS regulated and the US-DS control portfolios narrows. The decline in the market risk of the US-DS control portfolio (from the average value of about 0.8 to values around 0.3–0.4 towards the end of the period; see Fig. 4) is matched by a similar but slightly more moderate decline in the risk of the US-DS regulated portfolio.

In summary, we find that the changes identified in the risk of the regulated portfolio (Fig. 2) and the UK-DS regulated portfolio (Fig. 3) are not matched by changes in the risk of the US-DS regulated portfolio (Fig. 5) in the profit sharing period. That is, the changes we identify are not caused by changes in worldwide conditions.

Furthermore, Figs. 4 and 5 indicate robustness in the decrease in risk of the selected old-economy stocks in the 1999–2000 period. All six portfolios (drawn across the U.K. and

¹⁸The slight, insignificant deepening in the beta time path observed in the second half of 1998 is smoothed out when we remove August observations from the data. As the implications of the Asian and Russian financial crises at that time are not the core of this paper, we do not discuss this finding further. The deletion of August 1998 observations from the U.K. data does not change the results.

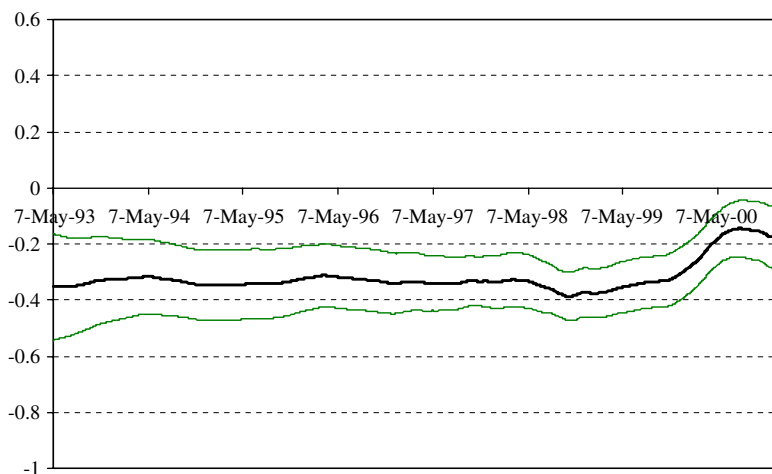


Fig. 5. Kalman filter estimates of the CAPM Market model beta coefficient for the difference in returns between the US-DS regulated and US-DS control portfolios over the period May 1, 1993–December 31, 2000 using daily observations. The thick line shows the coefficient and the two thin lines show the 95% confidence intervals.

U.S. companies) display a similar pattern of decline. Given that this paper is concerned with isolating and measuring the impact of changes in expectations of regulation in the U.K. during the profit sharing period, the implications of the paper do not depend on any impact in the final subperiod provided it is common across portfolios. However, the fact that the effects we find are consistent across all six portfolios strengthens the interpretation of the evidence (i.e., the third period is not a “freak” result of the U.K. sample). We conjecture that this decline in risk is likely to be the result of e-commerce effects during this period. We close this subsection with a brief discussion of why this might be the case. Note, the final decline in the market risk of the old-economy stocks in both the U.K. and the U.S. markets cannot be attributed to their overperformance of the local market indexes. The only group of shares that overperforms the corresponding index is the US-DS regulated portfolio in the third subperiod. Even if the sharp increase in the share price of the US-DS regulated portfolio in the third subperiod could drive the equity betas down, it would not explain an even larger decline in the market risk of the US-DS control portfolio and any of the U.K. portfolios. The same argument applies to the profit sharing period share performance.

While the impact on portfolio risk and return of high-tech companies in the period 1999 to 2000 is well documented, the cause, i.e., correct valuation versus bubble, is still the center of debate (see for example, Ofek and Richardson, 2003; Pastor and Veronesi, 2004, 2005; Schultz and Zaman, 2001; Schwert, 2002). Because the effect of e-commerce was most pronounced on the Nasdaq exchange, it would be no surprise if the betas of old-economy stocks would fell most heavily against such an index. However, all three of the market indexes (FTSE All Shares, UK-DS and US-DS) we employ in this study were also affected by the e-commerce bubble. Over the sample period these indexes were changing, with the old-economy stocks representing a smaller proportion of the market in 2000 than in 1993. Table 2 shows that regardless of which of the three approaches is considered, i.e., FTSE All Shares, UK-DS, or US-DS, the relevant regulated and control portfolios have

Table 2

Performance of indexes and portfolios (%). Columns denoted “DS regulated” and “DS control” denote statistics for the portfolios obtained using DS calculated indexes that contain companies included in the regulated and the control portfolios
 “DS total market” is the global market index calculated by DataStream. Statistics are based on daily observations.

Return	U.K.				U.S.			
	FTSE All Shares	Regulated portfolio	Control portfolio	DS total market	DS regulated	DS control	DS total market	DS regulated DS control
<i>Whole period (May 1, 1993–December 31, 2000)</i>								
Total	102.0	83.2	97.4	107.3	94.5	84.5	127.8	92.1 94.1
Av. monthly	1.1	0.9	1.1	1.2	1.0	0.9	1.4	1.0 1.0
Annualized	13.3	10.8	12.7	14.0	12.3	11.0	16.7	12.0 12.3
<i>Pre-profit sharing period (May 1, 1993–June 30, 1997)</i>								
Total	62.6	56.4	73.8	64.6	60.0	59.1	82.1	43.2 68.7
Av. monthly	1.3	1.1	1.5	1.3	1.2	1.2	1.6	0.9 1.4
Annualized	15.0	13.5	17.7	15.5	14.4	14.2	19.7	10.4 16.5
<i>Profit sharing period (July 1, 1997–August 12, 1999)</i>								
Total	34.3	32.0	26.2	37.2	34.7	26.5	46.0	37.3 24.2
Av. monthly	1.4	1.3	1.0	1.5	1.4	1.1	1.8	1.5 1.0
Annualized	16.5	15.3	12.6	17.9	16.7	12.7	22.1	17.9 11.6
<i>Post-profit sharing period (August 13, 1999–December 31, 2000)</i>								
Total	5.2	–5.2	–2.5	5.5	–0.3	–1.1	–0.3	11.6 1.1
Av. monthly	0.3	–0.3	–0.1	0.3	0.0	–0.1	0.0	0.7 0.1
Annualized	3.6	–3.7	–1.8	3.9	–0.2	–0.8	–0.2	8.2 0.8

Table 3

Correlations of daily returns with Nasdaq returns

“UK-DS total market” and “US-DS total market” are the global market indexes calculated by DataStream for the U.S. and U.K. markets, respectively.

Index	Pre-profit sharing period May 1, 1993–June 30, 1997	Post-profit sharing period August 13, 1999–Dec 31, 2000	Change (%)
US-DS total market	0.797	0.877	10
<i>Exchange rate adjusted</i>			
FTSE all shares	0.223	0.373	67
UK-DS total market	0.223	0.370	66
<i>No exchange rate adjustment</i>			
FTSE all shares	0.262	0.408	56
UK-DS total market	0.265	0.404	53

grown less than the market index. That is, price changes have been such that the companies within the regulated and control samples have experienced a relative decline overall. This decline holds before one considers the effect of changes in the composition of companies over the period. Therefore, the market indexes may have become more similar to Nasdaq over the sample period, and in particular, the daily returns may have become more correlated. Table 3 shows that indeed, this was the case. For the first and final subperiods Table 3 gives the correlations of daily returns (adjusted and not adjusted for the exchange rate) between Nasdaq and the market indexes we use.

The fact that the US-DS total market includes both Nasdaq and NYSE companies is one explanation for the high correlation of daily returns and the smaller change over time. The fact that the risk changes of the US-DS and the UK-DS control portfolios (Fig. 4) are very similar provides separate evidence that the U.K. market was affected by the e-commerce bubble (there is no real equivalent to Nasdaq in the U.K.). We believe that the changing nature of the indexes combined with the use of the Kalman filter on daily data that is responsible for the large decrease in risk at the end of the sample period. Note that the corresponding Nasdaq/NYSE return correlations are 0.733 in the opening subperiod and 0.688 for the final subperiod (i.e., correlation of daily returns declines). Furthermore, the pure correlation of the Nasdaq and NYSE indexes were 0.984 in the first subperiod and 0.181 in the final subperiod compared to 0.986 and 0.824 for the US-DS total market index we use in this subsection to derive U.S. risk.

5.2. Leverage

It is theoretically feasible that differences in the development over time of leverage ratios between the regulated and control portfolios could have been responsible for the evidence we attribute to the regulatory change. Note that because we consider the difference between the two portfolios, it is the relative change in leverage ratios that is relevant. Information on the total market value of the debt for the companies is unavailable. Many of the companies in the sample did not have debt that was traded over the full sample period, and even for those companies that had, such debt only formed a fraction of their

total debt. Therefore it is not possible to obtain the market value of the entire debt for these companies. Because we address the difference between portfolios, converting debt to market value for some companies is likely to be more misleading than using book value for all companies. Thus, in this subsection we first address the leverage effect using book values and we then consider the market price of debt to show that changes in leverage cannot be responsible for the diversion of betas between regulated and control portfolios during the profit sharing period.

To calculate asset betas for the portfolios, we use the formula $\beta_{\text{asset}} = (1 - l)\beta_{\text{equity}}$, where leverage $l = \text{long-term debt} / (\text{long-term debt} + \text{market value of equity})$. First, we calculate an average beta per calendar year from the Kalman filter regressions for each company in the control and regulated portfolios. We then use the annual leverage figures to convert these average annual equity betas into asset betas for each company. These are used to create betas of four equally weighted portfolios (the equity betas for the regulated and control portfolios, and the assets betas for these two portfolios). The differences between the equity and asset betas for the regulated and control portfolios are presented in Fig. 6. The trend identified using the Kalman filter regressions for the difference in equity betas (Figs. 2 and 3) is confirmed in the differences of the annual equity and asset betas. The correlation between the differences in equity betas and differences in asset betas is 0.99.

As indicated, while it is impossible to provide a full analysis of the market value of debt for all companies, for many companies there is a market price for some tranches of debt. In addressing this data it is useful to distinguish between two issues, namely, (i) the impact on our core result concerning the effect of regulation, and (ii) identification of any impact of the regulation on market debt prices.

- (i) The relative changes in asset betas across subperiods (and hence the core result) would only differ from those presented above if there were very large and systematic differences in the differences between the market and book values of debt between the

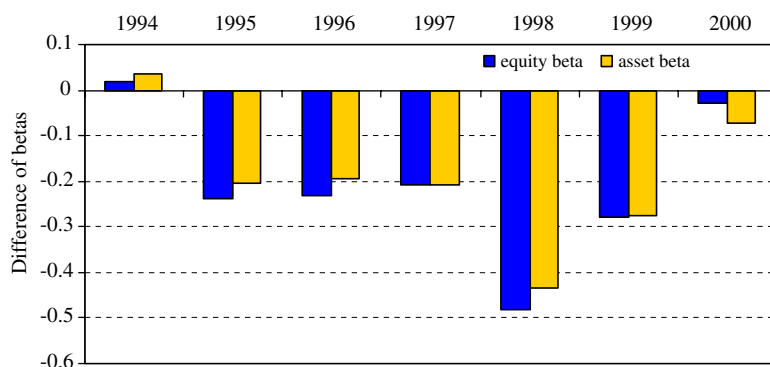


Fig. 6. Differences between betas of the regulated and control portfolios. The equity beta differences are based on daily company equity betas (derived from Kalman filter regressions) averaged for each calendar year. The asset beta differences are based on daily company asset betas (derived from the company annual equity betas using the formula $\beta_{\text{asset}} = (1 - l)\beta_{\text{equity}}$, where l is long-term debt/(long-term debt + market value of equity) averaged for each calendar year.

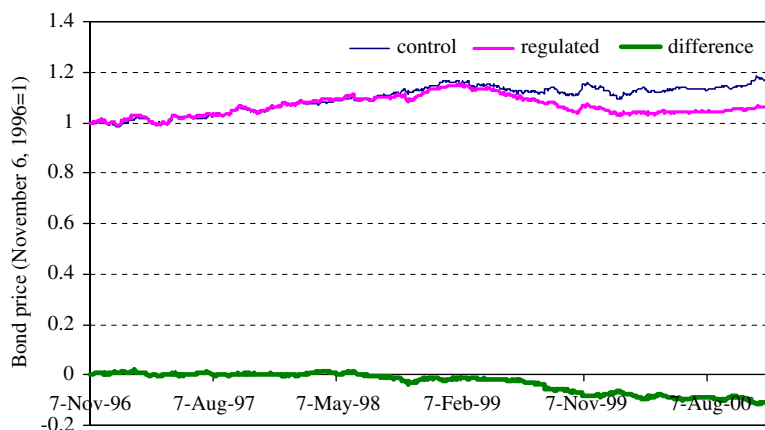


Fig. 7. Average bond price movement for equally weighted portfolios of bonds that were traded by companies in the control and regulated portfolios.

two portfolios. This is extremely unlikely. To verify this we create two portfolios (regulated and control) of bonds that were traded by companies in the portfolios. To obtain a relatively good representation of bond issues and to maximize the number of companies for which information is available prior to the profit sharing period, we start the sample on November 6, 1996. This allows us to collect information on the price movement of bonds for eight regulated companies (all together 19 time series) and 14 control companies (32 time series). More details can be found in Appendix D. Each price series is normalized to one on November 6, 1996, so both the regulated and control bond portfolios start at one (Time series in US\$ are converted into pound sterling). The time paths of equally weighted portfolios are given in Fig. 7.

It is evident that there is nearly no difference in the price of traded debt across the two portfolios during the profit sharing period. Thus, there is no suggestion from this evidence that the results obtained using book value leverage ratios are not robust.¹⁹

- (ii) The data on the market price of debt show that the two portfolios move together during the profit sharing period but the market price of debt for the regulated portfolios fell slightly relative to the market price of debt for the control portfolio. We expect that the similarity of the prices for the regulated and control portfolios in the profit sharing period is the result of two counteracting effects. First, the gearing is increasing for the regulated portfolios relative to the control portfolio. This should reduce the price of debt for the regulated portfolios relative to that of the control portfolio. Second, the proposed regulatory changes should make regulated debt more attractive. These two effects counteract each other. However, with the conclusion of the profit sharing period, the difference in leverage should predominate and the price of regulated debt should fall relative to that of the control portfolio. This may explain the effect observed in Fig. 7.

¹⁹Fig. 7 weighs each bond equally. We also weight bond portfolios per issuer to give each company equal representation in the average. The outcome is not sensitive to the form of aggregation.

Taken together, the calculated asset betas and the evidence provided on the market price of debt during the profit sharing period show that the regulatory changes we identify are robust.

6. Empirical analysis: FF 3F model

Fama and French (FF, 1992, 1993) enrich the CAPM model specification by introducing two additional factors, the difference between returns on portfolios of small and large stocks (the SMB factor) and the difference between returns on portfolios of high and low book-to-market ratios (the HML factor). This section investigates the FF3F model to verify whether it can offer additional insight.²⁰

As we indicate in the previous section, we have to use monthly data for the FF3F specification because the HML factor can only be constructed on a monthly basis.²¹ The regression equation estimated is

$$R_{i,t} = \alpha_{i,t} + \beta_{iM,t}R_{M,t} + \beta_{iSMB,t}R_{SMB,t} + \beta_{iHML,t}R_{HML,t} + \varepsilon_{it}, \quad (8)$$

where $R_{i,t}$ refers to return on the portfolio of regulated companies, the portfolio of the control portfolio, or the difference between these two portfolios at time t , $R_{M,t}$ is the return on the market portfolio (represented by the FTSE All Shares index), and $R_{SMB,t}$ and $R_{HML,t}$ refer to the two additional factors suggested by FF. Here the state vector is defined as $a_{i,t} = [\alpha_{i,t}, \beta_{iM,t}, \beta_{iSMB,t}, \beta_{iHML,t}]'$, where time $\eta_{i,t}$ is a 4×1 vector of serially uncorrelated disturbances with zero mean and covariance matrix Q . We assume that the covariance matrix is diagonal, which helps with the model's numerical tractability (the small number of observations makes it impossible to expand the covariance matrix structure). We believe that this simplification does not impact on the results because the unconditional correlation of the factors is low. The correlation of the market portfolio returns with the SMB (HML) factor is -0.21 (0.005); the correlation of the SMB and HML factors is -0.31 . In the case of OLS regressions for differences between the regulated and control portfolios on monthly observations, tests on residuals show that the Jarque-Bera test cannot reject the null hypothesis of normality (probability of 0.986) and that the White heteroskedasticity test cannot reject the null hypothesis of no heteroskedasticity (probability 0.194). Therefore, there is some justification for dropping the GARCH specification, especially since the calculations are cumbersome due to the limited number of observations. This simplification in the model specification reduces the number of parameters to be calculated for the Likelihood function from eight, if the GARCH structure of the error term is adopted, to just five.

Our primary aim is to see whether and to what degree the results presented in Section 4 are sensitive to the SMB and HML factors. To investigate this, as in the previous section, we run time-varying regressions according to Eq. (8) using monthly observations. Later we use OLS WHC, because the White test is positive, i.e., heteroskedasticity is present, for several individual companies.

We estimate the time paths of all four coefficients in Eq. (8) for the regulated portfolio, for the control portfolio, and for the difference of returns between the portfolios at each

²⁰Several papers are critical about the significance of FF's findings (see, e.g., Campbell and Vuolteenaho, 2003). Chung et al. (2006) suggest that the two additional FF factors are proxies for higher order co-moments.

²¹We would like to thank Alan Gregory for providing SMB and HML factors (used in Gregory et al., 2003).

point in time. However, for brevity we only present the time paths of the difference between returns of the regulated and control portfolios, as these have the most explanatory power in our analysis.²² Fig. 8 presents time paths (thick lines) and their corresponding 95% confidence intervals (thin lines) for the market factor, the SMB factor, and the HML factor.

The results for the market factor closely replicate the pattern previously estimated using higher frequency data for the single-factor model. Although there is weaker convergence in the post-profit sharing period relative to the single-factor model, the upward trend back towards the previous level of risk is very clear. The estimated value of the constant term is close to zero (i.e., -0.000088) as expected and is statistically insignificant from zero (the 95% confidence bounds are -0.009 and 0.009).

The other two factors do not seem to provide extra explanatory power. While the HML factor displays a very similar pattern to the market portfolio, given its standard error, it is not possible to conclude that the time path is significantly different from zero for the whole period in question. In contrast, the SMB factor is virtually flat. We do not present time paths estimated for the FF3F model for the regulated portfolio separately; however, neither of the time paths estimated for the two additional FF factors are significantly different from zero.

The smaller number of data points associated with the monthly data series we use in this section may be partly responsible for the outcomes we observe. Changing the data frequency modifies the standard errors. In particular, going from higher to lower frequency data increases standard errors by $\sqrt{n}$, where n is the difference in the relative number of observations per period, e.g., $n = 20$ when moving from daily to monthly observations. Although the shape of the time paths presented in Figs. 2 and 8 are very similar, the absolute values of the estimates and the corresponding errors differ. The 95% confidence intervals presented in Fig. 8 are approximately five times those presented in Fig. 2, which is consistent with the predicted increase by a factor of $\sqrt{20} \approx 4.47$. While the level of the betas differ between Figs. 2 and 8, the difference between the two paths is not statistically significant. Overall, we are able to conclude from the Kalman filter analysis that the pattern displayed by the market portfolio is strongest and is consistent with the theoretical predictions.

Before moving on to the time-invariant regression analysis of the FF3F model, it is helpful to estimate the market model on the monthly data. Table 4 presents results of the OLS WHC regression, analogous to those shown in Table 1. The signs of the coefficients and their significance are preserved. Also, the monthly estimates for the profit sharing dummy are larger than their daily counterparts, but the differences between these two are not significant.

Table 4 presents the OLS WHC results when the difference between the returns on the regulated and control portfolios, and the returns on the regulated portfolio, are used as a dependent variable both using single-factor and FF3F models with the profit sharing dummy. Table 4 also shows the OLS WHC results of the FF3F model specification with the profit sharing dummy using the difference between the returns on individual companies and the returns of the control portfolio.

²²The other six graphs, i.e., for the regulated and the control portfolios, can be obtained from the authors upon request.

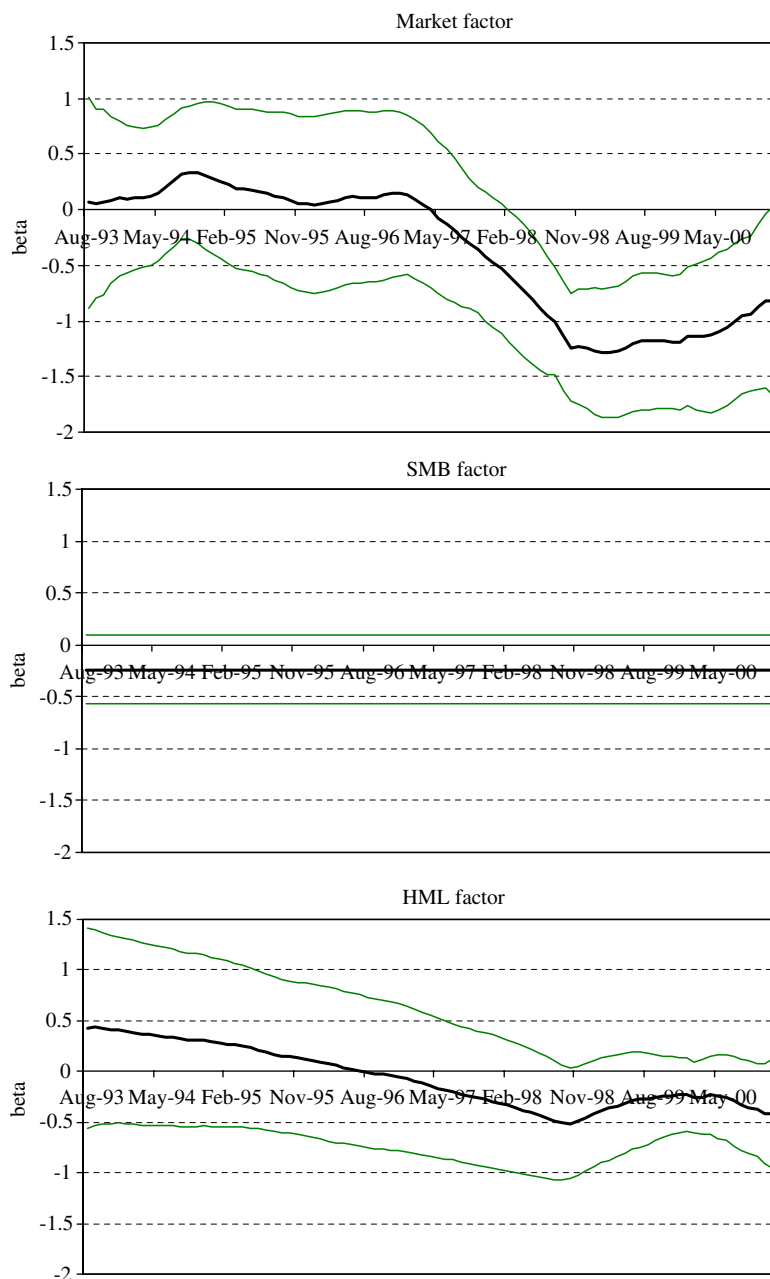


Fig. 8. Kalman filter estimates of the Fama-French three-factor model for the difference in returns between the regulated and control portfolios over the period May 1, 1993–December 31, 2000 using monthly observations. The thick line shows the coefficient and the two thin lines show the 95% confidence intervals.

We find that all three dummies for the FF factors are significant. The profit sharing dummy for the market factor is significant at the 1% level, for the SMB factor at the 5% level, and for the HML factor at the 10% level for the difference of the portfolios, and at

Table 4

Regression results for the single-factor and the three-factor Fama-French models

The β_M , β_{SMB} , and β_{HML} denote coefficients associated with the market, SMB, and HML portfolios, respectively. The α coefficient represents the constant term. The corresponding profit sharing dummy coefficients for each of the factors and for the constant carry the “ps” subscript. The profit sharing dummy is equal to one between July 1, 1997 and August 12, 1999, and zero otherwise. In the “Individual company level” panel the dependent variables are defined as the difference between returns on individual company (indicated by its name) and the return on the control portfolio. Regressions are based on monthly observations covering the May 1, 1993–December 31, 2000 period.

$R_{reg,t}$	α	α_{ps}	β_M	$\beta_{M,ps}$	β_{SMB}	$\beta_{SMB,ps}$	β_{HML}	$\beta_{HML,ps}$
<i>Aggregate level</i>								
$R_{reg,t} - R_{c,t}$	0.000 (0.005)	0.009 (0.012)	-0.144 (0.199)	-0.831** (0.387)				
$R_{reg,t}$	0.002 (0.004)	0.004 (0.009)	0.634*** (0.203)	-0.474* (0.283)				
$R_{reg,t} - R_{c,t}$	0.000 (0.005)	0.007 (0.011)	-0.113 (0.207)	-0.977*** (0.375)	0.210 (0.247)	-0.825** (0.367)	0.062 (0.191)	-0.870* (0.495)
$R_{reg,t}$	0.001 (0.004)	0.003 (0.009)	0.676*** (0.178)	-0.599** (0.289)	0.235 (0.200)	-0.644** (0.279)	0.4452*** (0.156)	-0.756** (0.335)
<i>Individual company level</i>								
Anglian Water	-0.003 (0.008)	0.010 (0.015)	-0.101 (0.259)	-1.260*** (0.419)	0.086 (0.375)	-0.702 (0.505)	0.293 (0.290)	-0.553 (0.772)
BAA	0.000 (0.005)	0.000 (0.017)	-0.062 (0.155)	-0.256 (0.325)	0.299 (0.271)	-0.359 (0.542)	0.429** (0.168)	-0.969* (0.497)
BT	-0.018** (0.009)	0.041*** (0.018)	0.581** (0.289)	-1.514*** (0.550)	-0.926*** (0.342)	0.143 (0.643)	-0.649* (0.335)	-0.281 (0.662)
Hyder	-0.007 (0.008)	-0.003 (0.015)	-0.251 (0.317)	-0.843* (0.485)	0.023 (0.372)	-0.608 (0.513)	0.602* (0.324)	-1.529** (0.720)
Kelda	0.000 (0.011)	0.005 (0.019)	0.001 (0.374)	-1.118** (0.562)	-0.055 (0.450)	-0.632 (0.583)	0.293 (0.404)	-1.058 (0.798)
National Grid	0.007 (0.009)	0.009 (0.016)	-0.227 (0.266)	-1.021** (0.441)	-0.532 (0.335)	-0.478 (0.489)	-0.709*** (0.243)	-0.268 (0.658)
National Power	0.006 (0.011)	-0.017 (0.017)	-0.079 (0.367)	-0.739* (0.454)	0.725 (0.569)	-1.075* (0.634)	-0.321 (0.374)	-0.963 (0.59)
Pennon	-0.006 (0.010)	0.023 (0.016)	-0.455 (0.330)	-0.731 (0.497)	0.720* (0.516)	-1.253** (0.611)	0.772*** (0.274)	-1.416** (0.673)
PowerGen	0.010 (0.009)	-0.018 (0.020)	-0.042 (0.316)	-1.083** (0.507)	1.250* (0.642)	-1.921** (0.756)	0.489 (0.330)	-1.773*** (0.664)
Scottish Power	0.003 (0.008)	0.003 (0.015)	-0.079 (0.264)	-0.842** (0.414)	0.018 (0.275)	-0.845* (0.455)	-0.245 (0.280)	-0.408 (0.544)
Scottish & Southern Energy	0.001 (0.009)	0.011 (0.020)	-0.019 (0.318)	-0.906 (0.563)	0.368 (0.424)	-1.045* (0.632)	0.246 (0.427)	-1.027 (0.929)
Severn Trent	0.001 (0.009)	0.007 (0.018)	-0.269 (0.322)	-1.159** (0.551)	0.518 (0.440)	-1.005* (0.579)	0.419 (0.296)	-1.015 (0.630)
Thames Water	0.011 (0.013)	0.001 (0.019)	-0.654 (0.674)	-0.772 (0.812)	0.546* (0.313)	-1.179** (0.512)	0.189 (0.240)	-0.597 (0.615)
United Utilities	-0.004 (0.007)	0.011 (0.014)	0.032 (0.275)	-1.234*** (0.423)	0.050 (0.248)	-0.497 (0.373)	-0.224 (0.222)	-0.678 (0.591)
Viridian	0.008 (0.009)	0.009 (0.018)	0.130 (0.308)	-1.380** (0.536)	-0.070 (0.337)	-0.787 (0.522)	-0.656** (0.408)	-0.509 (0.662)

OLS WHC indicate that the estimates were obtained by OLS with White heteroskedasticity consistent standard errors and covariance corrections.

Standard errors are in parentheses. *, ** and *** denote significance at the 10%, 5%, and 1% level, respectively.

the 5% level for the regulated portfolio. Here the strength of the market factor relative to the other factors, found in the Kalman filter estimates, is confirmed, but now the other factors display significance since the imposition of the assumption of constant betas (save for the profit sharing period) eases the requirements on the data. We also note that the market factor is more significant in the FF3F specification than in the pure market model.

The company-specific data offer less clear results. We believe the main reason for this is that at this level, the company specific variability combined with monthly frequency data makes it hard to pick up effects due to regulatory changes. Because the portfolio of regulated companies smooths these company-specific effects, the results are stronger for this portfolio. However, at the company level we find evidence that is consistent with the portfolio results (i.e., strong results for the market factor dummy, slightly weaker results for the SML dummy, and significant but weak results for the HML dummy). The profit sharing dummy for the market factor is significant for 11 of the companies, the profit sharing dummy for the SMB factor is significant for seven companies, and the profit sharing dummy for the HML factor is significant for four companies.

Taken together, the results from the FF3F model confirm the view of Section 4 that there is a consistent profit sharing effect. Thus while, the primary aim of the paper is to assess the impact of regulation on risk, the results also provide a contribution to the literature on the significance of the FF3F.

7. Conclusions

We identify that the existing literature does not give clear evidence relating to how the time-varying nature of risk responds to expected regulatory changes designed to impact on risk. We believe that the scarcity of such evidence is because, although regulatory changes typically do impact on risk, there is a lack of relatively clean cases that isolate the risk effect for investigation. This paper provides a detailed study of a case in which regulation was introduced specifically to change the risk transfer between parties, but not to create a wealth transfer. In the late 1990s the U.K. government proposed a change in the policy relating to prices of all regulated U.K. utilities, namely, from a price-cap to a system of explicit profit sharing between companies and customers. We apply Kalman filter techniques, OLS with White error corrections, and ML with GARCH effects to a single-factor market model and the FF3F model to identify risk changes.

Using daily data with a market model the effect is clear on time-varying betas for a portfolio of regulated companies. This is true whether we apply the Kalman filter technique to the full sample period or to three subsamples (before, during, and after the treatment). The effect is also clear when applying the Kalman filter to individual companies. Using either OLS with White error correction or ML with GARCH effects, we find that the effect on risk is significant at the 1% level on a portfolio of regulated companies, and on almost every company at the individual-company level. We use U.S. data to confirm that we do not identify a worldwide effect. We also address the issue of leverage changes and show that they do not explain differences between the regulated and control portfolios.

We also investigate the effect of the impact of the regulatory change on risk using the FF3F model on monthly data. Here the effect is confirmed although the two additional factors appear to contribute less explanatory power than the market factor. The Kalman filter approach only confirms the effect as significant through the market factor. However,

once we impose the assumption of constant beta (save for the profit sharing period) the requirements on the data are eased. Applying OLS to the portfolios shows that the profit sharing dummies are significant for all three factors. This is true, though less so when we address individual company-level data. As pointed out in the introduction, the case study provides a tangential but intriguing contribution to the debate on the role of the FF3F.

Taking the single-factor and three-factor evidence together, the paper shows that the impact on risk of regulatory changes specifically designed to impact on risk is both significant and consistent with theory.

Appendix A

The content of the regulated and control portfolios of companies are given on Table A.1.

Table A.1

List of companies included in the regulated and control portfolios

Regulated portfolio		Control portfolio	
Company	Main business area	Company	Main business area
Anglian Water	Water and Sewerage Company	Allied Domecq	Distillers
BAA	Airport Services	Blue Circle Industries	Industrial materials
BT	Telecommunications	BOC Group	Chemicals
Hyder	Water and Sewerage Company	Boots	Drug stores
Kelda	Water and Sewerage Company	BP Amoco	Oil refiners & distribution
National Grid Company	Electricity Transmission and Telecommunications	British Aerospace	Aircraft manufacture
National Power	Electricity Generation & Supply	British Airways	Airlines
Pennon	Water and Sewerage Company	Cadbury Schwepps	Confectionery goods
PowerGen	Electricity Generation & Supply	Diageo	Distillers
Scottish Power	Electricity Generation & Supply	GKN	Metal product manufacturers
Scottish & Southern Energy	Electricity Generation & Supply	Glaxo Wellcome	Pharmaceuticals
Severn Trent	Water and Sewerage Company	Granada Compass	Leisure services
Thames Water	Water and Sewerage Company	Imperial Chemical Industries	Chemical manufacture
United Utilities	Water and Sewerage Company	Marks & Spencer	Clothes/food retail
Viridian	Electricity Generation & Supply	P & O Steam Navigation	Sea transport
		Smith Kline Beecham	Pharmaceuticals
		Tate & Lyle	Sugar production
		Tesco	Food retail

FTSE30 companies excluded from the control sample:

Regulated (BG Group, British Telecom, Scottish Power).

Financial services (Lloyds TSB, Prudential Corporation, Royal & Sun Insurance, Royal Bank Of Scotland).

Electronics (Invensys, Marconi).

Information, Communication, and Media (Emi Group, Reuters, Vodafone Airtouch).

Source: www.ftse.com/ft30.txt & www.corporateinformation.com

Appendix B

Kalman filter estimates of the CAPM market model are given in Fig. B.1.

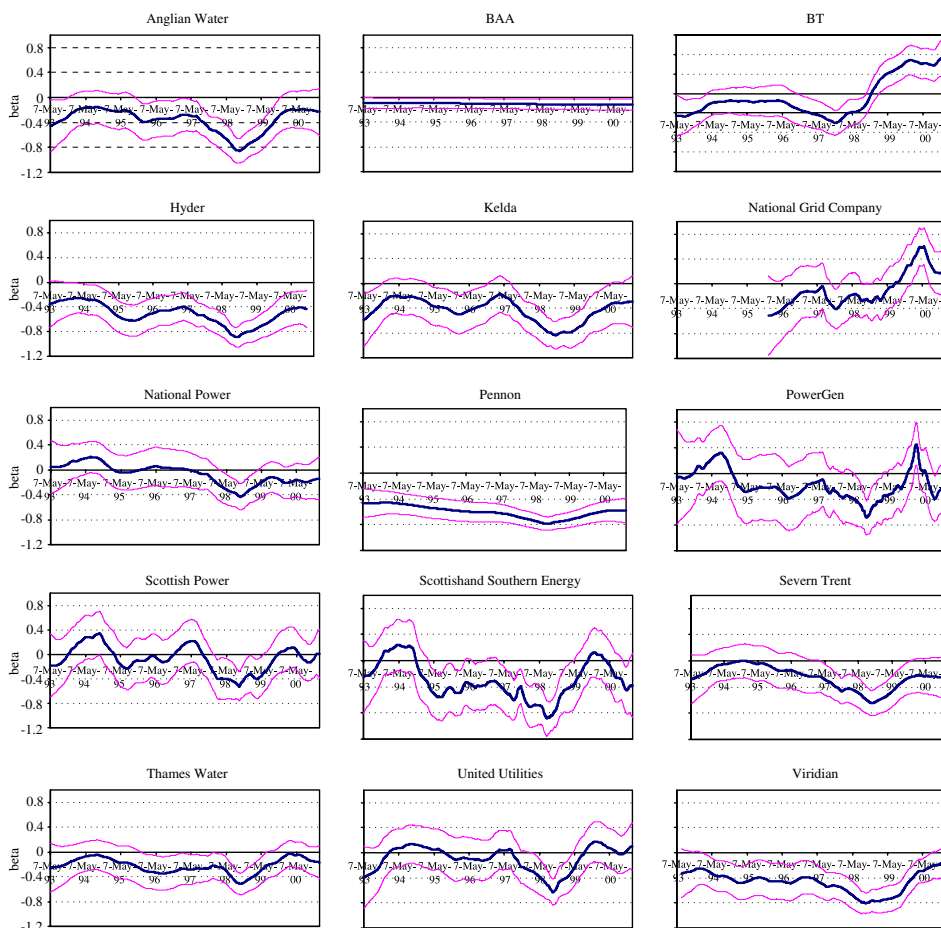


Fig. B.1. Kalman filter estimates of the CAPM Market model beta coefficient for the difference between individual regulated companies (indicated by name) and the control portfolios over the period May 1, 1993–December 31, 2000 using daily observations. The market portfolio is represented by the FTSE All Share index. The thick lines show the coefficient and the thin lines show the 95% confidence intervals.

Appendix C

Time paths of the beta coefficient are given in Fig. C.1.

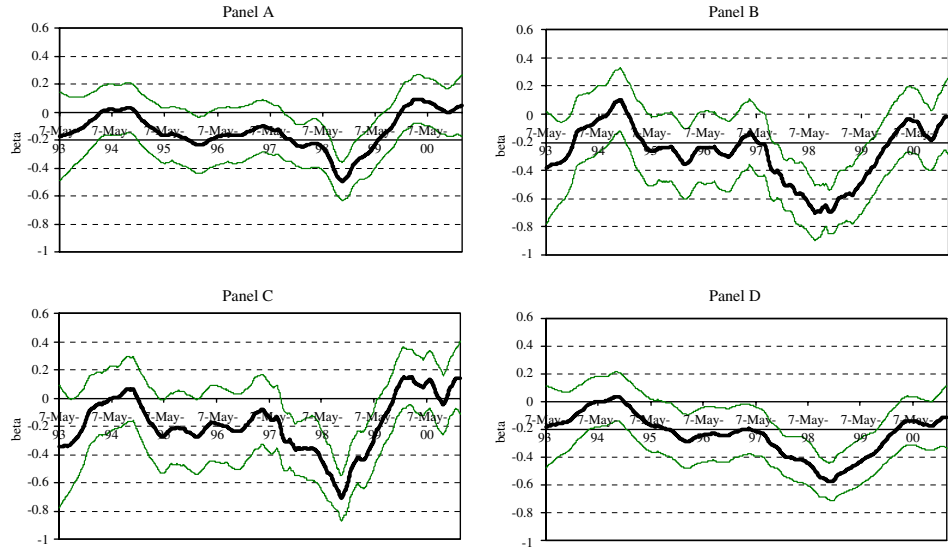


Fig. C.1. Time paths of the beta coefficient estimated for the difference in returns between the regulated portfolio and subsamples of the control portfolio. Panels A and B correspond to the control portfolio subsamples consisting of the 9 smallest and of the 9 largest companies, respectively. Panels C and D correspond to the first and the second half of the control portfolio, respectively, when ranked alphabetically.

Appendix D

Bond prices of regulated and control companies are given in Table D.1.

Table D.1
Regulated and control companies for which bond prices are analyzed

Regulated companies		Control companies	
Name	Number of bonds	Name	Number of bonds
Anglian Water	3	Blue Circle Industries	1
BAA	3	BOC Group	3
BT	5	Boots	1
Hyder	2	BP Amoco	2
National Grid Company	2	British Aerospace	1
PowerGen	1	British Airways	2
Severn Trent	2	Cadbury Schwepps	2
Thames Water	1	Glaxo Wellcome	3
		Granada Compass	2
		Imperial Chemical Industries	1
		P&O Steam Navigation.	4
		Smith Kline Beecham	5
		Tate & Lyle	2
		Tesco	3

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